

Function Calculator with Partial Derivatives

Problem Description

You are asked to implement a calculator that evaluates nested function compositions using the following predefined functions:

1. $f(x) = 5x^2 + 6x + 4$
2. $g(x, y) = 6x^2 + 3x + 8y^2 - 15y$
3. $dg/dx(x, y) = 12x + 3$
4. $dg/dy(x, y) = 16y - 15$
5. $h(x, y, z) = x + y - z$

The program should read a single expression consisting of nested function calls and compute the outputs value, for example, if the input is $h(f(2), dg/dx(3,4), dg/dy(5,6))$, your program should output -6.000.

Input	Output
$h(f(2), dg/dx(3,4), dg/dy(5,6))$	-6.000

You must implement using **yacc (bison)** and **lex (flex)**.

Input Format

- Each input consists of a single line containing one expression.
- Parameters can be
 - ◆ integers or floating-point numbers (in decimal notation, e.g., -1.2, 2.3)
 - ◆ results of other function calls
- If the input is syntactically invalid or semantically incorrect (e.g., wrong number of arguments)

the program must print: **Invalid**

Output Format and Precision

- For a valid expression, output the computed result as a floating-point number.
- The result must be printed with exactly three decimal places, using standard rounding (四捨五入).
- **No** newline at the end.
- Hint: `printf("%.3f\n", xxx)`

Tokens Definition

- `FFUNC ::= "f"`
- `GFUNC ::= "g"`
- `HFUNC ::= "h"`
- `DGDX ::= "dg/dx"`
- `DGDY ::= "dg/dy"`
- `NUM ::= -?[0-9]+([0-9]+)?`
- `LPAREN ::= "("`
- `RPAREN ::= ")"`
- `COMMA ::= ","`

Grammar Specification

- `startsymbol ::= function`
- `function ::= Unary_call | Binary_call | Ternary_call`
- `Unary_call ::= Unary_op LPAREN expression RPAREN`
- `Binary_call ::= Binary_op LPAREN expression COMMA expression RPAREN`
- `Ternary_call ::= Ternary_op LPAREN expression COMMA expression COMMA expression RPAREN`
- `Unary_op ::= FFUNC`
- `Binary_op ::= GFUNC | DGDX | DGDY`
- `Ternary_op ::= HFUNC`
- `expression ::= function | NUM`

Sample Inputs and Outputs

Input	Output
<code>h(f(2),dg/dx(10,5),dg/dy(20,0))</code>	174.000
<code>f(dg/dx(2,3))</code>	3811.000
<code>g(h(f(4),dg/dx(0,50),dg/dy(7,5)),2)</code>	12836.000
<code>h(1.123453236,2,3)</code>	0.123
<code>g(3)</code>	Invalid

Note: If `fprintf()` does not output normally, you can try using `printf()` instead.